

SAGAR INSTITUTE OF RESEARCH AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MINOR PROJECT : CS-506

Project Title

”AI Driven MCQ Quiz Generator (Shuffle Based) ”

Team Members :

- 1. Pranshu Mishra**
- 2. Priyaranjan Singh**

Under the Supervision of :

Prof. Amit Sengar

LIST OF CONTENTS

- Introduction
- Problem Statement
- Objective
- Proposed Methodology
- Abstract
- Literature Survey
- Execution Plan
- Application of the proposed work
- Conclusion
- References



INTRODUCTION

Today's students rely heavily on quick practice tools for exam preparation, especially multiple-choice questions. But manually creating MCQs is time-consuming, repetitive, and often lacks variation.

The **AI Driven MCQ Quiz Generator** solves this by using artificial intelligence to automatically generate fresh, unique, and topic-relevant MCQs within seconds.

This system helps students revise faster, teachers prepare tests efficiently, and ensures that learning becomes more interactive, adaptive, and personalized.



PROBLEM STATEMENT

- Teachers spend unnecessary time creating MCQs manually.
- Students depend on limited fixed-question practice sets.
- No automatic system exists to generate topic-wise, difficulty-based MCQs instantly.
- Many quiz platforms repeat the same old questions, reducing learning effectiveness.
- There is a need for a tool that creates new, diverse, and accurate MCQs automatically using AI.



OBJECTIVE

- To design an AI-based system that automatically generates MCQs from any given topic or text.
- To provide students with unlimited practice questions with different difficulty levels.
- To help teachers create quizzes, assignments, and tests quickly.
- To generate accurate options, correct answers, and explanations using AI.
- To develop a fast, user-friendly, and reliable quiz-generation platform.



PROPOSED METHODOLOGY

1. Input Layer

User selects a subject, topic, or uploads study text.

2. AI Processing Layer

AI model analyzes the content → extracts key concepts → forms MCQs.

- Question generation
- Option creation
- Correct answer identification
- Explanation creation

3. Quiz Generation Layer

System prepares a complete MCQ set with:

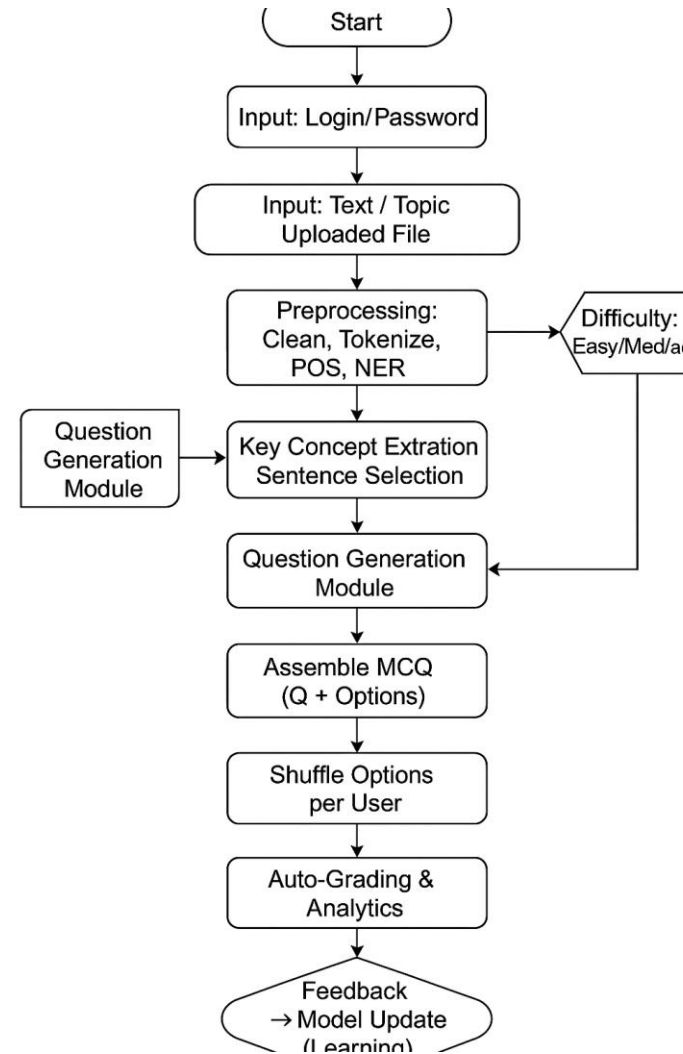
- Questions
- Options
- Answer key
- Explanations

4. Output Layer

MCQs shown to the user or exported as PDF/quiz file.



TECHNICAL ARCHITECTURE



ABSTRACT

The **AI Driven MCQ Quiz Generator** is a smart system designed to automatically generate multiple-choice questions using artificial intelligence. It analyzes input text or topic keywords and produces high-quality MCQs with correct answers and explanations. This helps both students and teachers by saving time and increasing learning efficiency. The system aims to modernize education by providing unlimited, diverse, and accurate quiz content.



LITERATURE REVIEW

Year	Reference	Result	Limitations	Conclusion
2024	Shalini & Priyadharshini, Automated Question Generation (IEEE)	NLP + DL generate MCQs effectively	Distractors sometimes irrelevant; high model cost	Transformer-based QG effective but needs refinement
2023	Soni & Singh, AI-Based Exam System (IJRET)	MCQ generation + shuffle per user	No adaptive difficulty; shuffle limited	Improve fairness; future adaptive models
2022	Bhattacharya et al., BERT Quiz Generator (Springer)	High-accuracy MCQs using BERT.	Heavy computation, complex grammar	Domain-tuned models needed



2021	Ahmed & Kumar, AI Question Generator (IJCTT)	Auto MCQ + shuffled options	Limited question types	Add analytics + descriptive support
2020	Chaware et al., AI-NLP MCQ Generator (IRJET)	Keyword extraction + synonyms	Predictable options, no difficulty level	Strong base; requires advanced NLP

EXECUTION PLAN

Phase 1: Foundation & Design

- Define clear workflows for chat, intervention, and counselor booking
- Create user personas to design a friendly and comfortable UI
- Build Flutter screens with wellness-friendly color psychology
- Create database schema for users, sessions, and counselor calendars

Phase 2: Backend Core Development

- Implement API endpoints using FastAPI
- Add JWT-based secure authentication
- Build appointment scheduling logic
- Implement rules for emergency escalation when a student shows critical distress signals



EXECUTION PLAN

Phase 1: Requirement & Design

- UI design
- Topic selection page
- Database structure

Phase 2: Backend Development

- API creation
- MCQ generation logic
- AI model integration

Phase 3: AI Integration

- Implement question-generation model
- Add difficulty-level logic
- Test MCQ accuracy

Phase 4: Testing & Finalization

- Validate generated questions
- Improve UI
- Prepare final deployment



APPLICATION OF THE PROPOSED WORK

- Students → Practice unlimited MCQs.
- Teachers → Create quick tests and assignments.
- Coaching Institutes → Generate large question banks.
- Competitive exam aspirants → Fresh questions for practice.
- Colleges → Use for internal quizzes, labs, and exams.



CONCLUSION

The **AI Driven MCQ Quiz Generator** provides a modern, fast, and intelligent way to produce high-quality MCQs. It saves time, reduces manual effort, and ensures fresh question sets every time. The system supports both teachers and students, making learning more interactive, adaptive, and efficient. This project demonstrates how AI can transform the traditional approach to question creation in education.



REFERENCES

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Thank You

